AI-Powered Industrial Safety Intelligence Platform

for Zero-Harm Operations

Sahyadri Petrochem & Specialty Chemicals Ltd. (SPSCL) — Simulated Facility

Detailed Project Document — Phase II Submission

ET AI Hackathon 2026 — **Problem Statement #1:** AI-Powered Industrial Safety Intelligence for Zero-Harm Operations

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github.com/pixel-pudding/Industrial-Safety-Intelligence

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1. Executive Summary

Indian heavy industry does not have a sensor problem — it has an intelligence problem. Plants already have SCADA, gas detectors, permit-to-work systems, and CCTV. What they lack is a layer that fuses these independent systems into a single risk picture and acts before a fatality, not after. This is the exact gap the 2025 Visakhapatnam Steel Plant coke-oven explosion exposed: functioning sensors, no connected reasoning.

We built a working AI-Powered Industrial Safety Intelligence Platform for a simulated petrochemical facility (SPSCL) that implements all six agent types suggested by the problem statement, not a narrowed subset: a Compound Risk Detection Engine, Digital Permit Intelligence Agent, Incident Pattern Intelligence (RAG), Emergency Response Orchestrator, Geospatial Safety Heatmap, and Quality & Compliance Audit Agent. Every prediction carries evidence, confidence, and a citation — no black-box output.

The system runs against a scripted Industrial Digital Twin (10 demo scenarios, culminating in a flagship compound cascade mirroring the fatal-incident pattern cited in the problem statement), a live dashboard with a Digital Twin, AI reasoning panel, and Safety Copilot, and a real-time WebSocket data layer connecting them.

2. Problem Context

India recorded over 6,500 fatal workplace accidents in FY2023 (DGFASLI), excluding most mining and construction. At Visakhapatnam Steel Plant in January 2025, eight workers died when entrapped gases triggered a coke-oven explosion — despite functioning gas detectors, permit-to-work controls, and SCADA. Investigators found the warning signals existed; no intelligence layer connected them to a decision in time.

A 2024 FICCI survey found over 60% of large industrial facilities still coordinate between their own digital safety tools through manual handoffs. The technology exists. The unified reasoning layer over it does not.

3. Our Solution — Guiding Principles

- The AI is the product; the dashboard is only a visualization layer.
- The AI correlates multiple independent signals to detect compound risk conditions no single sensor would flag alone.
- The AI recommends concrete interventions, not just alerts.
- Every prediction includes evidence, reasoning, confidence, and supporting data — no black-box answers.
- The plant is assumed already digitally modernized (SCADA, IoT, CCTV, ePTW, CMMS, worker tracking) — this platform is an intelligence layer sitting above those systems, not a replacement for them.
- Everything is simulated except the safety manuals, SOPs, and regulatory guidance, which are real public documents (OISD, Factories Act 1948) used for retrieval-augmented generation.

4. Simulated Facility: Sahyadri Petrochem & Specialty Chemicals Ltd.

A fictional but grounded petrochemical facility in the Panvel–Raigad industrial belt, Maharashtra, producing Ethylene Oxide/Ethylene Glycol, chlor-alkali (caustic soda + chlorine), industrial solvents, and downstream specialty surfactants/resins. Two distinct hazard classes — explosive/flammable and toxic — coexisting on one campus allow genuinely cross-hazard compound-risk scenarios, grounded in real Indian chemical-corridor precedent.

10-Zone Facility Layout

Zone	Purpose	Key Data	Compound-Risk Relationship
A	Reactor Block	EO/EG production	Fed by Zone F cooling; can propagate to Zone C overpressure
B	Chlor-Alkali Unit	Electrolysis (chlorine + caustic)	Independent hazard class; fed by Zone F power/cooling
C	Tank Farm	EO/solvent storage	Bridge zone between A and D; receives product from A
D	Solvent Blending & Packaging	Blend/package toluene, MEK	Primary source of permit + maintenance scenarios
E	Surfactant/Resin Plant	Downstream EO conversion	Deliberate low-hazard control zone
F	Utilities Block	Boilers, compressors, cooling towers	Single point of failure affecting multiple zones
G	Loading Bay	Tanker loading	External-party risk (drivers, static discharge)
H	Maintenance Workshop	CMMS + permit issuance	Upstream source of maintenance-overdue flags site-wide
I	Control Room/Admin	SCADA aggregation, shift handover	Consumption point for all agents
J	Main Gate/Security	Access control, CCTV hub	Source of worker-location and permit-context data
K	Flare Stack & N2 Inerting	Emergency venting, purging	Silent failure point — visible only cross-checked with permits

Hazard Profile & Failure Modes

Two distinct hazard classes coexist on one campus: explosive/flammable (EO reaction, solvent vapors) and toxic (chlorine). This is deliberate — it is what makes genuinely cross-hazard compound scenarios possible, and it mirrors real Indian chemical-corridor plants (Maharashtra/Gujarat) where multiple hazard classes routinely sit within one facility boundary.

Zone	Possible Hazards	Failure Modes	Safety Requirements
A	Explosion/runaway, burns	Cooling failure, catalyst degradation	Continuous monitoring, cooling redundancy
B	Toxic gas release	Compressor failure, seal degradation	Gas detection, PPE, evacuation plan
C	Overpressure, vapor explosion	Relief valve failure, overfill	Interlocks, vapor detection, hot-work restrictions
D	Fire, explosion	Static discharge, hot work near vapor	Grounding checks, gas testing before permits
E	Thermal (low severity)	Dryer overheat	Standard monitoring
F	Cascading utility failure	Cooling fouling, compressor overload	Redundant cooling, pressure relief
G	Static discharge, spill	Grounding/coupling failure	Grounding checks, containment
H	Indirect (originates elsewhere)	Backlog accumulation	CMMS compliance tracking
J	Unauthorized access	Check-in bypass	Escort requirements, access logging
K	Uncontrolled venting	Ignition failure, skipped purge	Mandatory purge verification

Permit-to-Work System

Permits are validated at issuance only in a conventional ePTW system. The core design gap this platform targets is detecting when conditions drift dangerous after approval — something existing permit systems structurally cannot see.

Permit Type	Required When	Compound-Risk Interaction
Hot Work	Welding/cutting/grinding	Cross-checked live against VOC/CL2 readings — conditions can drift after approval
Confined Space Entry	Entry into reactors/tanks	Should require N2-purge completion first — a gap here is a key detection target
Electrical Work	Live/isolated electrical work	Interacts with electrical load/voltage tags
Working at Height	Elevated tank/structure work	Interacts with evacuation planning
General Work	Routine non-hazardous tasks	Control case — should not be flagged

5. System Architecture

Approach: independent specialized agents with a central orchestrator, rather than one monolithic reasoning core. This aligns directly with the problem statement's "Agentic AI / Multi-Agent Systems" suggestion and is required by the demo scenarios themselves — Scenario 10, for instance, needs a genuine Detection → Incident Retrieval → Emergency Response handoff, not one model doing everything.

The Six Agents

Agent	Triggered By	What It Does
Compound Risk Detection Engine	Continuous, every SCADA/IoT tick	Scores 7 independent evidence categories (sensor, cross-zone utility, maintenance, permit, worker presence, CCTV, shift handover) — never a single-threshold alert
Digital Permit Intelligence Agent	Permit issuance + continuous re-check	Cross-checks all 6 permit types against live plant conditions; flags approve / flag / block-recommend with cited reasons
Incident Pattern Intelligence (RAG)	Called at Warning tier by the Detection Engine	Retrieves matching historical incidents + regulatory excerpts (OISD, Factories Act) from ChromaDB, with confidence bands, not blind certainty
Emergency Response Orchestrator	Called at Critical tier	Evacuation cascade, multi-channel alert dispatch, frozen evidence snapshot, preliminary regulatory-style report
Geospatial Safety Heatmap	Continuous	Live zone-by-zone risk view combining risk tier, worker count, active permits, hazard classification
Quality & Compliance Audit Agent	Scheduled + anomaly-triggered	Continuously checks CMMS/permit records against OISD/Factory Act, flags compliance gaps before an audit would

Orchestration Model

Hybrid hub-and-spoke: the Compound Risk Detection Engine is the hub, calling Incident Pattern Intelligence and Emergency Response directly when its own score crosses tier thresholds. Permit Intelligence, the Heatmap, and Compliance Audit run semi-independently and feed back into the Detection Engine's scoring. All agents share one handoff contract, a risk event object: zone, timestamp, risk_score, tier, contributing_signals, triggered_agents. A fully-meshed "every agent calls every agent" design was deliberately avoided — harder to debug and harder to diagram clearly for evaluation.

Technology Stack

Layer	Choice	Why
Agent orchestration	LangGraph	Models the hub-and-spoke pattern directly
LLM provider	Google Gemini (Flash, free tier)	Sustained free tier with no card requirement, unlike alternatives
Backend/API	FastAPI (Python)	Async-friendly for continuous simulated SCADA ticks
RAG vector store	ChromaDB (embedded)	Zero infrastructure setup
Structured data	SQL (SQLAlchemy ORM)	Relational fit for SCADA/CMMS/permit schemas
Real-time layer	Python asyncio tick loop + WebSockets	Powers the live Industrial Digital Twin
Frontend	React + Tailwind, Vite	Digital Twin, AI reasoning panel, live heatmap, Safety Copilot

6. Data Foundation & Retrieval-Augmented Generation (RAG)

10 structured historical incident records (each with hazard category, contributing factors, severity, and a free-text narrative) plus real regulatory excerpts — OISD-STD-105 and Factories Act 1948 (§36 confined space, §87 dangerous operations) — are loaded into ChromaDB. The Incident Pattern Intelligence agent structurally pre-filters by zone/hazard category before semantic retrieval, which measurably improves precision over pure vector search alone, and every retrieved match carries a confidence band (high/moderate/low) so the reasoning layer treats a weak precedent honestly rather than as certainty.

7. Industrial Digital Twin & Demo Scenarios

A manually-triggered scenario engine (not randomly auto-fired, so scenarios can be cleanly recorded and re-recorded) ramps specific sensor tags from Normal to Warning to Critical over 30–60 seconds — long enough to demonstrate a real lead-time advantage, never an instant jump. Ten scripted scenarios validate the system end to end:

#	Scenario	Demonstrates
1	Single sensor breach (RX-TEMP → Warning)	Baseline alerting
2	Single breach, quick resolution	Avoids over-alarming on transients
3	Overdue maintenance alone	Confirms no false alert from this signal alone
4	Overdue maintenance + Warning reading	First compound signal
5	Hot-work permit approved safe, drifts dangerous	Conditions-changed-after-approval gap
6	Chlorine compressor vibration + workers nearby	Cross-hazard-type reasoning
7	Cooling tower degradation affecting Zones A & B	Clearest “no single sensor sees this” case
8	Skipped N2 purge + confined space request	Proactive permit blocking
9	Shift changeover: flagged condition not escalated	Temporal reasoning across shifts
10	Full compound cascade (5 independent sources)	Flagship scenario — mirrors the fatal-incident pattern

8. Dashboard & User Experience

The dashboard is deliberately not a wall of every possible metric. Each screen is built to answer three questions: what is happening, why is it happening, and what should I do. Core modules: an interactive Digital Twin (the visual centerpiece), an AI Risk Panel with full reasoning/evidence/confidence/citation, a live Incident Timeline, Recommended Interventions, a Geospatial Heatmap, and an AI Safety Copilot for natural-language queries against the same live agent outputs.

9. Alignment to Judging Criteria

Criterion	Weight	How This Project Addresses It
Innovation	25%	Rule-based compound scoring across 7 independent evidence categories (not single-sensor thresholds), with an LLM reasoning layer invoked only at Warning/Critical to stay within free-tier limits — a deliberate cost/accuracy tradeoff, not a shortcut.
Business Impact	25%	Directly modeled on the Visakhapatnam coke-oven pattern the problem statement cites — sensors existed, no intelligence layer connected them in time. Every agent output traces to real regulatory text (OISD, Factories Act) and a structured historical-incident corpus, not fabricated numbers.
Technical Excellence	20%	6 independently testable agents, a documented handoff contract (risk event object), deterministic replay verified across scenario re-runs, and honestly disclosed limitations (e.g., RAG rank-1 precision noise with only 10 incidents) rather than overstated accuracy.
Scalability	15%	Hub-and-spoke orchestration (vs. fully-meshed agent calls) chosen specifically for clarity and debuggability at scale; schema and agent boundaries designed to add zones/sensors/permit types without redesigning the core engine.
User Experience	15%	Dashboard answers the three questions every screen should: what is happening, why, what to do — Digital Twin, AI Reasoning, Recommended Interventions, and a Safety Copilot, without burying the judge in every possible metric at once.

10. Engineering Rigor & Honest Limitations

We verified deterministic replay across scenario re-runs (identical output on repeated triggers of the same scenario, after fixing state-reset bugs found during testing), and disclose rather than hide known constraints: with only 10 seeded incidents and a generic embedding model, RAG rank-1 precision among closely related incidents is noisier than production-scale retrieval would be, and Quality & Compliance Audit citations are lower-confidence for maintenance-interval gaps than for hot-work/confined-space gaps, since the regulatory corpus was scoped toward the latter. Both are flagged as future-work items.

11. Roadmap

- Replace scripted CCTV events with real computer-vision inference on camera feeds.
- Expand the historical-incident and regulatory corpus beyond the 10-incident seed set to reduce RAG retrieval noise.
- Add DGMS-specific regulatory coverage for facilities with a mining component.
- Move from SQLite/simulated SCADA to a real OPC-UA/SCADA integration for pilot deployment.

12. Submission Details

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